

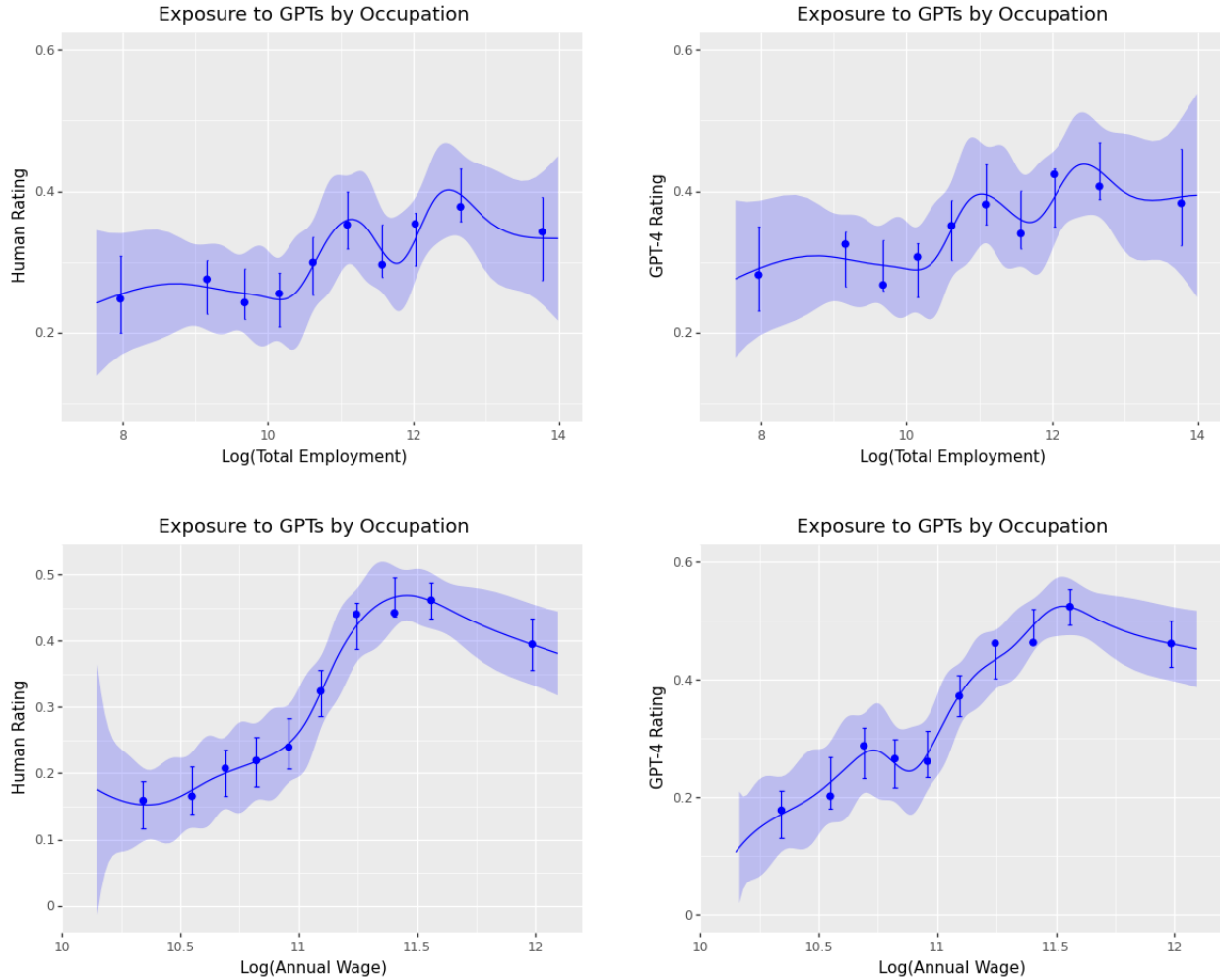


Figure 4: The binscatter plots depict the exposure to language models (LLMs) in various occupations, as assessed by both human evaluators and GPT-4. These plots compare the exposure to LLM and partial LLM-powered software (β) at the occupation level against the log of total employment within an occupation and log of the median annual wage for occupations. While some discrepancies exist, both human and GPT-4 assessments indicate that higher wage occupations tend to be more exposed to LLMs. Additionally, numerous lower wage occupations demonstrate high exposure based on our rubric. Core tasks receive twice the weight of supplemental tasks within occupations when calculating average exposure scores. Employment and wage data are sourced from the BLS-OES survey conducted in May 2021. In aggregating tasks to the occupation-level, we assign half the weight to O*NET supplemental tasks as we do for core tasks. All weights within an occupation sum to one.

4.3 Skill Importance

In this section, we explore the relationship between the importance of a skill for an occupation (as annotated in the O*NET dataset) and our exposure measures. First, we use the Basic Skills provided by O*NET (skill definitions can be found in Appendix B) and normalize the measure of skill importance for each occupation to improve the comprehensibility of the results. Next, we conduct a regression analysis on our exposure measures (α , β , ζ) to examine the strength of associations between skill importance and exposure.

Our findings indicate that the importance of **science** and **critical thinking** skills are strongly negatively associated with exposure, suggesting that occupations requiring these skills are less likely to be impacted by current LLMs. Conversely, **programming** and **writing** skills show a strong positive association with exposure, implying that occupations involving these skills are more susceptible to being influenced by LLMs (see Table 5 for detailed results).

4.4 Barriers to Entry

Next, we examine barriers to entry to better understand if there is differentiation in exposure due to types of jobs. One such proxy is an O*NET occupation-level descriptor called the "Job Zone." A Job Zone groups occupations that are similar in (a) the level of education needed to get a job in the occupation, (b) the amount of related experience required to do the work, and (c) the extent of on-the-job training needed to do the work. In the O*NET database, there are 5 Job Zones, with Job Zone 1 requiring the least amount of preparation (3 months) and Job Zone 5 requiring the most extensive amount of preparation, 4 or more years. We observe that median income increases monotonically across Job Zones as the level of preparation needed also increases, with the median worker in Job Zone 1 earning \$30,230 and the median worker in Job Zone 5 earning \$80,980.

All of our measures (α , β , and ζ) show an identical pattern, that is, exposure increases from Job Zone 1 to Job Zone 4, and either remains similar or decreases at Job Zone 5. Similar to Figure 3, in Figure 5, we plot the percentage of workers at every threshold of exposure. We find that, on average, the percentage of workers in occupations with greater than 50% β exposure in Job Zones 1 through 5 have β at 0.00% (Job Zone 1), 6.11% (Job Zone 2), 10.57% (Job Zone 3), 34.5% (Job Zone 4), and 26.45% (Job Zone 5), respectively.⁸

4.4.1 Typical Education Needed for Entry

Since inclusion in a Job Zone accounts for both the education required—which itself is a proxy for skill acquisition—and the preparation required, we seek data to disentangle these variables. We use two variables from the Bureau of Labor Statistics' Occupational data: "Typical Education Needed for Entry" and "On-the-job Training Required to Attain Competency" in an occupation. By examining these factors, we aim to uncover trends with potential implications for the workforce. There are 3,504,000 workers for whom we lack data on education and on-the-job training requirements, and they are therefore excluded from the summary tables.

Our analysis suggests that individuals holding Bachelor's, Master's, and professional degrees are more exposed to LLMs and LLM-powered software than those without formal educational credentials (see Table 7). Interestingly, we also find that individuals with some college education but no degree exhibit a high level of exposure to LLMs and LLM-powered software. Upon examining the table displaying barriers to entry, we observe that the jobs with the least exposure require the most training, potentially offering a lower payoff (in terms of median income) once competency is achieved. Conversely, jobs with no on-the-job training required or only internship/residency required appear to yield higher income but are more exposed to LLMs.

⁸For this set of results, all tasks have equal weight within an occupation. Results do not change meaningfully with the core/supplemental weighting scheme.

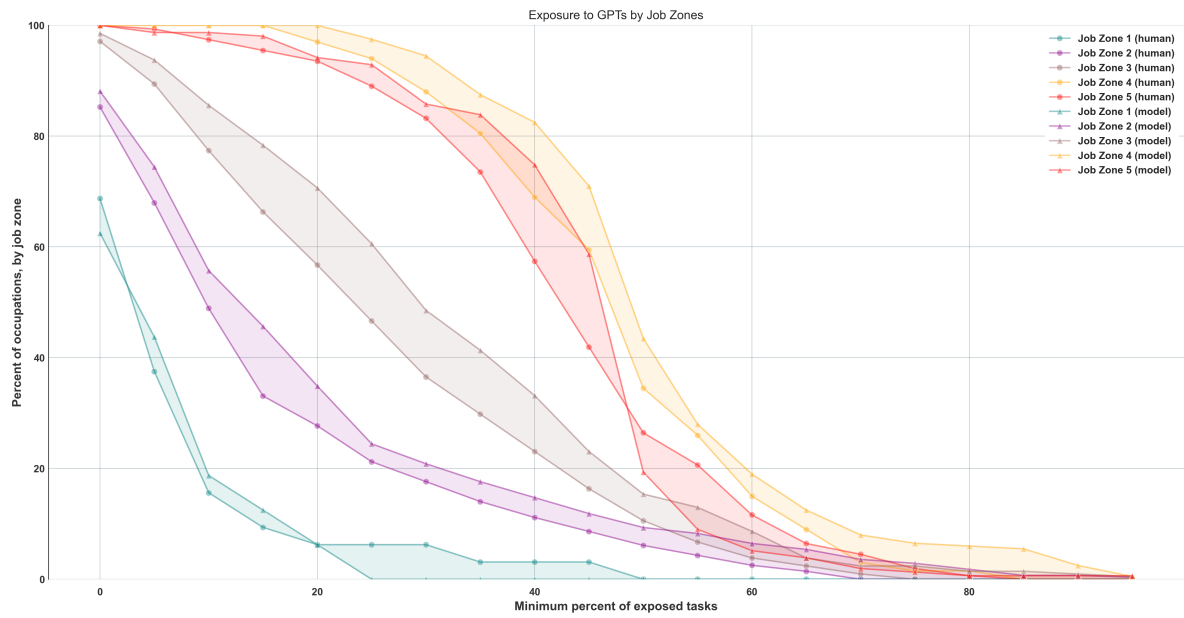


Figure 5: β exposure ratings of occupations in the five Job Zones, which are groups of similar occupations that are classified according to the level of education, experience, and on-the-job training needed to perform them. All tasks are weighted equally.

Group	Occupations with highest exposure	% Exposure
Human α	Interpreters and Translators	76.5
	Survey Researchers	75.0
	Poets, Lyricists and Creative Writers	68.8
	Animal Scientists	66.7
	Public Relations Specialists	66.7
Human β	Survey Researchers	84.4
	Writers and Authors	82.5
	Interpreters and Translators	82.4
	Public Relations Specialists	80.6
	Animal Scientists	77.8
Human ζ	Mathematicians	100.0
	Tax Preparers	100.0
	Financial Quantitative Analysts	100.0
	Writers and Authors	100.0
	Web and Digital Interface Designers	100.0
	<i>Humans labeled 15 occupations as "fully exposed."</i>	
Model α	Mathematicians	100.0
	Correspondence Clerks	95.2
	Blockchain Engineers	94.1
	Court Reporters and Simultaneous Captioners	92.9
	Proofreaders and Copy Markers	90.9
Model β	Mathematicians	100.0
	Blockchain Engineers	97.1
	Court Reporters and Simultaneous Captioners	96.4
	Proofreaders and Copy Markers	95.5
	Correspondence Clerks	95.2
Model ζ	Accountants and Auditors	100.0
	News Analysts, Reporters, and Journalists	100.0
	Legal Secretaries and Administrative Assistants	100.0
	Clinical Data Managers	100.0
	Climate Change Policy Analysts	100.0
	<i>The model labeled 86 occupations as "fully exposed."</i>	
Highest variance	Search Marketing Strategists	14.5
	Graphic Designers	13.4
	Investment Fund Managers	13.0
	Financial Managers	13.0
	Insurance Appraisers, Auto Damage	12.6

Table 4: Occupations with the highest exposure according to each measurement. The final row lists the occupations with the highest σ^2 value, indicating that they had the most variability in exposure scores. Exposure percentages indicate the share of an occupation's task that are exposed to GPTs (α) or GPT-powered software (β and ζ), where exposure is defined as driving a reduction in time it takes to complete the task by at least 50% (see exposure rubric A.1). As such, occupations listed in this table are those where we estimate that GPTs and GPT-powered software are able to save workers a significant amount of time completing a large share of their tasks, but it does not necessarily suggest that their tasks can be fully automated by these technologies. All tasks are assigned equal weight within an occupation.